Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of New South Wales (UNSW, Sydney)
Short name / acronym used in projects	UNSW, Sydney (NSW)
Participant Identification Code (PIC), if available	999862809
Website(s)	www.unsw.edu.au/medicine-health/our-schools/optometry
Country	Australia
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Level 5, Rupert Myers Building, North Wing, Gate 14, Barker St, UNSW, Sydney, 2052
Department(s), institute(s), research group(s) or unit(s) that would be involved	Ocular Surface Research Group; Eye Research Group; Retinal Disease and Clinical Imaging Research Group; Retinal Research Group; Ocular Pathology Research Group; and Optical Imaging and Visualisation (OIV) Lab.
Additional institutional resources relevant to collaboration	Mark Wainwright Analytical Centre (MWAC): https://www.unsw.edu.au/research/facilities-and-infrastructure/find-a-facility/mwac Ramaciotti Centre for Genomics: https://www.unsw.edu.au/science/ramaciotti 3D Culture Facility: https://www.unsw.edu.au/research/facilities-and-infrastructure/find-a-facility/3dcf

Main contact person for this profile	A/Prof Nicole Carnt, Director of Research, School of Optometry and Vision Science, UNSW, Sydney
Main contact email	n.carnt@unsw.edu.au
Other contact person(s), if relevant	A/Prof Maria Markoulli, Co-Director of Teaching, School of Optometry and Vision Science, UNSW, Sydney

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
x Ocular surface	x Retina
☐ Inherited retinal dystrophies (IRDs)	x Age-related ocular disease
x Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
x Advanced therapies / regenerative medicine	x Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	☐ Ocular vascular disease
x Infection / trauma-related ocular disease	x Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	x AI-assisted diagnosis / software / digital tools
x Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	corneal organoids; ocular surface disease; ocular allergy; tear biomarkers; retinal biomarkers; oculomics; multimodal ocular imaging; artificial intelligence; clinical trials; ocular infection
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
The participating UNSW teams undertake multidisciplinary translational vision science across ocular surface disease, retinal disorders, ocular pathology, advanced imaging, artificial intelligence, biomarker discovery, regenerative medicine and public health. The overarching research line is to advance the prevention, diagnosis and treatment of ocular disease by integrating discovery science, clinically relevant models, enabling technologies and patient-centred validation. Research strengths include ocular infection, inflammation and allergy; dry eye disease and contact lens-related ocular health; tear, tissue and imaging biomarkers; ocular pain; retinal degeneration and age-related macular degeneration; and therapeutic discovery and evaluation. Core platforms include human donor tissues, human corneal organoids and advanced 3D ocular tissue models, tissue explants, cell culture and animal models, genomics and molecular profiling, clinical cohorts, multimodal ocular imaging and clinical trials. Emerging oculomics research applies AI-enabled analysis of ocular imaging and retinal biomarkers to investigate ocular disease and associations with systemic health. Across these areas, the teams connect mechanistic discovery with biomarker and technology development, clinical validation and patient-centred research, supporting more precise diagnostics, therapeutics and prevention strategies.

Why your team could be relevant to a translational eye-health and vision consortium
UNSW offers a collaborative translational ecosystem that integrates discovery science, advanced disease modelling, biomarker development, clinical research and patient-centred validation. Complementary expertise across ocular surface disease, retinal disorders, ocular pathology, multimodal imaging, artificial intelligence and regenerative medicine enables research questions to be addressed from mechanism through to clinical evaluation.

Our added value lies in linking human donor tissues, human corneal organoids and advanced 3D ocular tissue models, preclinical models, multi-omics, clinical cohorts, imaging and trials. Collaboration across academia, healthcare, industry, regulatory stakeholders and health consumers helps align research priorities and translation pathways with scientific evidence, clinical need and lived experience. Health consumers can contribute to priority setting, study design, interpretation, dissemination and implementation.

Beyond ophthalmology, the teams contribute expertise in immunology, microbiology, neuroscience, ageing, bioinformatics, data science, oculomics and public health. This breadth supports interdisciplinary staff mobility, reciprocal skills transfer and new approaches at the interface of eye health and systemic health.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- Ocular surface disease, infection, inflammation, dry eye disease and ocular allergy
- Retinal disease mechanisms, neurodegeneration, ageing and neuroprotection
- Human donor tissue, ocular pathology and tissue-explant research
- Human corneal organoids and advanced 3D ocular tissue models
- Cell culture, preclinical and animal models for disease mechanisms and therapeutic evaluation
- Tear, tissue, molecular and imaging biomarker discovery and validation
- Genomics, transcriptomics, multi-omics and bioinformatics
- Quantitative multimodal ocular imaging, hyperspectral retinal imaging and retinal biomarker development
- Oculomics and AI-enabled analysis for ocular and systemic health research
- Clinical cohorts, clinical trials, patient-reported outcomes and translational validation
- Health consumer engagement, co-design and patient-centred research
- Drug discovery, screening and therapeutic development

Infrastructure, equipment, facilities or datasets relevant to consortium work

- UNSW Mark Wainwright Analytical Centre (MWAC): shared microscopy, imaging, mass spectrometry, flow cytometry and complementary analytical capabilities
- UNSW Ramaciotti Centre for Genomics: next-generation sequencing, gene-expression and transcriptomic analysis, single-cell approaches and bioinformatics
- UNSW 3D Culture Facility: organoid, explant and advanced cell-culture platforms for disease modelling and therapeutic discovery
- Advanced anterior and posterior ocular imaging facilities, including hyperspectral retinal imaging and Spectralis OCT/OCTA platforms
- Specialised ocular surface, retinal and ocular pathology laboratories, with access to human donor ocular tissues through collaborating eye banks
- Clinical cohorts in ocular surface disease, infection, allergy, dry eye disease, retinal disorders and age-related eye disease
- Biobanked tear, tissue, imaging and multi-omics datasets, with image-analysis, AI and bioinformatics workflows
- Clinical trial capability and health consumer networks supporting patient-centred validation

Training capacity and knowledge-transfer potential

Inbound staff could undertake reciprocal scientific, technical, clinical and translational training across:

- Human donor tissue, ocular pathology, tissue explants, human corneal organoids and advanced 3D ocular tissue models
- Cell culture, preclinical and animal models of ocular disease
- Tear and tissue biomarker collection, molecular profiling, genomics, transcriptomics and bioinformatics
- Multimodal ocular imaging, hyperspectral retinal imaging, OCT/OCTA, quantitative image analysis and oculomics
- AI and machine learning for imaging, biomarker discovery and clinical decision support
- Clinical research methods, trial design, patient-reported outcomes and translational validation
- Health consumer engagement, co-design, priority setting and patient-centred research
- Research ethics, governance and regulatory considerations relevant to clinical and translational studies
- Grant development, project management, scientific communication, research leadership and international collaboration

What your team would hope to gain from participating
Building on successful informal staff exchanges and international collaborations over the past five years, which have contributed to Q1 journal publications, higher degree research theses, research training and leadership development, the consortium would provide a structured mechanism to expand staff mobility, shared training and joint supervision. We seek access to complementary disease models, advanced therapeutic and regenerative-medicine platforms, patient-derived validation cohorts, multi-omics and data-science methods, and specialised imaging and computational infrastructure. We also aim to strengthen expertise in gene and cell therapies, neuroprotection, drug delivery, bioengineering, biomaterials, regulatory science, health technology assessment, implementation research and commercialisation. Engagement with academic, clinical, industry and consumer partners would support cross-sector exposure, patient-oriented validation and more effective translation. The intended outcomes are sustainable international partnerships, joint publications and supervision, collaborative funding, stronger research capability and longer-term pathways from discovery to clinical and societal benefit.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Nicole Carnt	Associate Professor/Director of Research	Ocular Surface Research Group	Academic	Mid	
Maria Markoulli	Associate Professor Co-Director of Teaching	Ocular Surface Research Group	Academic	Mid	
Fiona Stapleton	Scientia Professor/Clinical Teaching Director, UNSW	Eye Research Group	Academic	Senior	
Jackie Tan	Associate Professor Director Eye Research Group	Eye Research Group	Academic	Mid	
Mark Willcox	Professor	Eye Research Group	Academic	Senior	
Isabelle Jalbert	Professor	Ocular Surface Research Group	Academic	Senior	
Suki Jaiswal	Post Doctoral Scientist	Ocular Surface Research Group	Academic	Early	
Lisa Nivison-Smith	Senior Lecturer	The Retinal Disease and Clinical Imaging Research Group	Academic	Mid	
Daisy Shu	Senior Lecturer	Retinal Research Group	Academic	Early	
Michele Madigan	Associate Professor	Ocular Pathology Group	Academic	Senior	
Jerome Ozkan	Senior Lecturer	Ocular Surface Research Group	Academic	Mid	
Maitreyee Roy	Associate Professor	Ocular Imaging and Visualisation Group	Academic	Mid	

Main disciplines represented in your team	Optometry; vision science; cell and molecular biology; immunology; microbiology; ocular pathology; genomics and bioinformatics; biomaterials and tissue engineering; clinical research; data science and artificial intelligence; public health; and regulatory science.

Experience in EU / international / multi-partner collaborative projects	Experience in international and multi-partner collaborations across academia, healthcare and industry includes participation in COST Action CA24112, Implementation of Tear Fluid Biomarkers in Precision Medicine, and partnership in the FLAIR consortium proposal under Horizon Europe call HORIZON-HLTH-2025-01-TOOL-03 on multimodal generative AI in biomedical research.
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Clinical cohorts, industry-sponsored trials, patient-reported outcomes, health consumer engagement and patient-centred validation across ocular surface and retinal disease.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. Shrestha GS; Vijay AK; Stapleton F; White A; Pickford R; Carnt N, 2023, 'Human tear metabolites associated with nucleoside-signalling pathways in bacterial keratitis', *Experimental Eye Research*, 228, pp. 109409, http://dx.doi.org/10.1016/j.exer.2023.109409

2. Khan S; Markoulli M; Nureen L; Di Girolamo N; Willcox M, 2026, 'Effects of Sitagliptin and Celery Seed Extract on Corneal Nerve Morphology and Sensory Dysfunction in Diabetic Rats', *Nutrients*, 18, http://dx.doi.org/10.3390/nu18142243

3. Phu J; Khuu SK; Nivison-Smith L; Kalloniatis M, 2025, 'Standard automated perimetry for glaucoma and diseases of the retina and visual pathways: Current and future perspectives', *Progress in Retinal and Eye Research*, 104, http://dx.doi.org/10.1016/j.preteyeres.2024.101307

4. McLellan FC; Liang YG; Jin Y; Madigan MC; Ng PQ; White AJR; Williams PA; Shu DY, 2026, 'Synergistic Cytokine Signaling Drives Angiofibrotic Gene Pathways in Primary Human Retinal Endothelial Cells', *American Journal of Pathology*, 196, pp. 2237 - 2255, http://dx.doi.org/10.1016/j.ajpath.2026.06.007

5. Sarwat S; Willcox M; Stapleton F; Roy M, 2026, 'Biophysical Interactions of Silicon Quantum Dots With Human Tears and Dry Eye Drops: A Langmuir Trough Study', *Translational Vision Science and Technology*, 15, http://dx.doi.org/10.1167/tvst.15.1.10

Up to 5 relevant previous projects, grants, networks or collaborations

Examples on the diversity and breadth of research projects, networks and collaboration are detailed below:

1. Advancing Corneal Infection Research with Human Organoids
- Funding agency: UNSW Global Research Impact Program (GRIP), Round 3 (2025)
- Role: Coordinator / Lead Investigator (CIA)
- Investigators: A/Prof Nicole Carnt (CIA), A/Prof Ashok Tiwari, Prof Virender Sangwan, Prof Mark Willcox, Dr Anna Ferguson
- International: Yes (Australia-India collaboration)
- Focus: Development of human corneal organoid models to advance translational research in corneal infection.

2. AZR-MD-001 for Meibomian Gland Dysfunction (MGD)
- Funding agency: Azura Ophthalmics Pty Ltd (Industry-sponsored clinical trial)
- Role: Clinical trial site / research partner
- Investigators: A/Prof Jackie Tan, Scientia Prof Fiona Stapleton
- International: Yes (industry networks spanning Australia, USA, UK, Japan and China)
- Focus: Vehicle-controlled clinical trial evaluating safety and tolerability of AZR-MD-001 for meibomian gland dysfunction.

3. Understanding and Managing the Impact of Smoke on the Eyes of Australian Bush Firefighters
- Funding agency: Natural Hazards Research Australia, Core Research Program
- Role: Research partner

- Investigators: Dr Suki Jaiswal, Prof Isabelle Jalbert
- National collaboration: Australia
- Focus: Environmental impacts on ocular surface health and strategies to reduce vision-related effects of smoke exposure.

4. From Fumarates to Microtubules: Novel Anti-angiogenic and Anti-fibrotic Therapies for AMD

- Funding agency: BrightFocus Foundation (USA), New Investigator Grant Program
- Role: Project lead
- Investigator: Dr. Daisy Shu (CIA)
- International: Yes (USA-Australia)
- Focus: Development of novel therapeutic approaches for age-related macular degeneration.

5. Understanding Inner Retinal Remodelling to Improve Clinical Outcomes for Patients with Age-Related Macular Degeneration

- Funding agency: National Health and Medical Research Council (NHMRC) Investigator Grant (2020-2025)
- Role: Project lead
- Investigator: Dr. Lisa Nivison-Smith (CIA)
- National collaboration: Australia
- Focus: Mechanisms of retinal remodelling and translation of findings to improve outcomes for patients with AMD.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
• Granted patent family for Antimicrobial Compounds (2016-079), with national-phase protection in the United States (Patent No. 10,988,444; granted 2021), Australia (granted 2022) and New Zealand (Patent No. 754200; granted 2025). Inventors: R. Kuppusamy, D. Black, N. Kumar and M.D.P. Willcox • Human corneal organoid, tissue-explant and advanced 3D ocular tissue-model platforms • Clinical trial capabilities and protocols for ocular surface disease, dry eye disease, contact lenses and meibomian gland dysfunction • Established tear biomarker collection, ocular surface phenotyping and patient-reported outcome methods • Clinical imaging datasets and AI-enabled analysis pipelines for detection, monitoring and risk stratification • Ocular surface and retinal disease cohorts with linked clinical, imaging, biological and patient-reported data • Health consumer engagement frameworks for co-design, priority setting, dissemination and translation

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers x Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Researchers could be onboarded through the Research Practicum pathway: https://www.unsw.edu.au/study/international-students/study-abroad#researchpracticum

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Translational disease models: human corneal organoids, tissue explants, cell culture systems and animal models for mechanistic research and therapeutic evaluation. Ocular biomarkers and oculomics: tear, tissue, retinal and imaging biomarkers for disease detection, monitoring and risk stratification. Advanced clinical imaging and AI: multimodal anterior and posterior ocular imaging, quantitative image analysis and AI-enabled diagnostic and predictive modelling.

Capabilities or expertise you would especially like to access through the consortium
• Advanced therapies, regenerative medicine and gene/cell therapy platforms • Novel organoid systems, patient-derived models and validation cohorts • Multi-omics, systems biology and computational biology • AI, digital health and clinical decision-support technologies • Drug delivery, biomaterials and bioengineering • Regulatory science, health technology assessment and implementation research • Industry partnerships, commercialisation pathways and innovation ecosystems • International patient networks, longitudinal cohorts and real-world evidence platforms

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

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Date completed: 2 Oct 2026

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